

Progress Seminar 1

On

DeepFake Shield: Multi-Modal GenAI Fraud Detection Pipeline

Guide:

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Introduction

The Rise of Multi-Modal Fraud

Generative AI tools can now construct highly realistic facial synthetics and clone voices within seconds, creating weaponized multi-modal digital duplicates.

- **Traditional Limit:** Existing detectors are unimodal (checking either image or audio alone), failing when attackers synchronize fake media perfectly.
- **The Cross-Modal Gap:** Missing correlations (e.g., lip-speech mismatch, spectral anomalies) remain undetected without unified pipelines.
- **Proposed Solution:** Deepfake Shield Multi analyzes the correlation between audio and video vectors, ensuring holistic security.



Literature Survey

Sr. No.	Paper Title and its Author	Details of Publication	Findings
1	A Multimodal Framework for DeepFake Detection Xu B., et al.	arXiv, Oct. 04, 2024	Combined VGG19 and ANN. Extracted novelty distances and LGCN features. Achieved 94% accuracy, but lacks sub-250 ms real-time latency verification on live streams.
2	Enhancing DeepFake Detection with Multi-Modal Transformers .Hombis	MDPI / ACM DL, 2025	Utilized ResNet50, LSTM, image accuracy, and CNN-LSTM (91%) audio. High False Positives for artifacts learning and low fusion strategy yields.
3	Multi-Modal DeepFake Detection Systems Survey Shreya, T., et al	Forensics Sci., March 15, 2024 Pg no. 142-165(Volume 4 , issue 1)	Shows unimodal group under high compression (below 61%). Recommends mid-level cross-modal fusion to counter low-quality data vulnerability.
4	Unified Multimodal Deepfake Detector Marce, J.	ICML, June 2025 Pg no. 1102-1115	Implements MTCNN for face extraction + EfficientNet. Reaches 87.04% accuracy with an EER of 12.6%. Suffers when synthetic voiceovers are perfectly aligned.

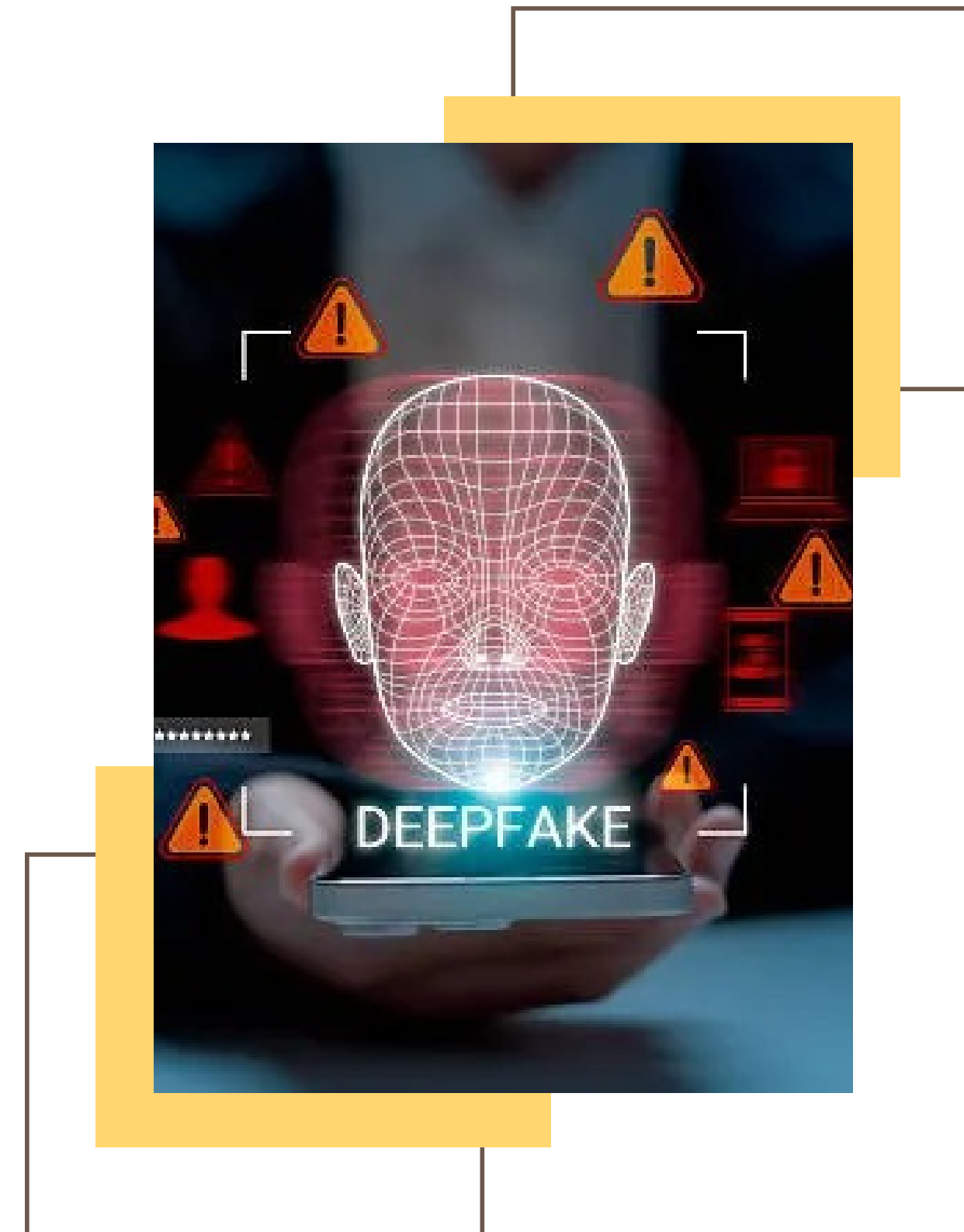
Objective of Project

- 01 Multi-Modal Feature Extraction**

Our first objective is to analyze different types of information, such as facial movements, lip movements, and voice. By che
- 02 Optimize Performance under Compression**

Our second objective is to make the system work effectively even when the video quality is reduced or the network is unstable. We aim to maintain high detection accuracy even in such conditions.
- 03 Feature Vector Alignment & Fusion**

Our third objective is to compare the audio with the person's lip movements. For example, if the voice says one thing but the lip movement does not match properly, the system can identify it as a possible deepfake.



Work Flow



PROCESS

1

Input Ingestion

Captures live streaming or uploaded video files. Splits audio track and video frames dynamically.

2

Feature Extraction

CNN Extracts image features. CNN + LSTM Analyzes video frames and motion patterns. Wav2Vec2 Extracts speech features from audio. BERT Extracts contextual features from text.

3

Cross-Modal Fusion

Self-attention blocks align visual lip features with speech pitch, tone, and prosody vectors.

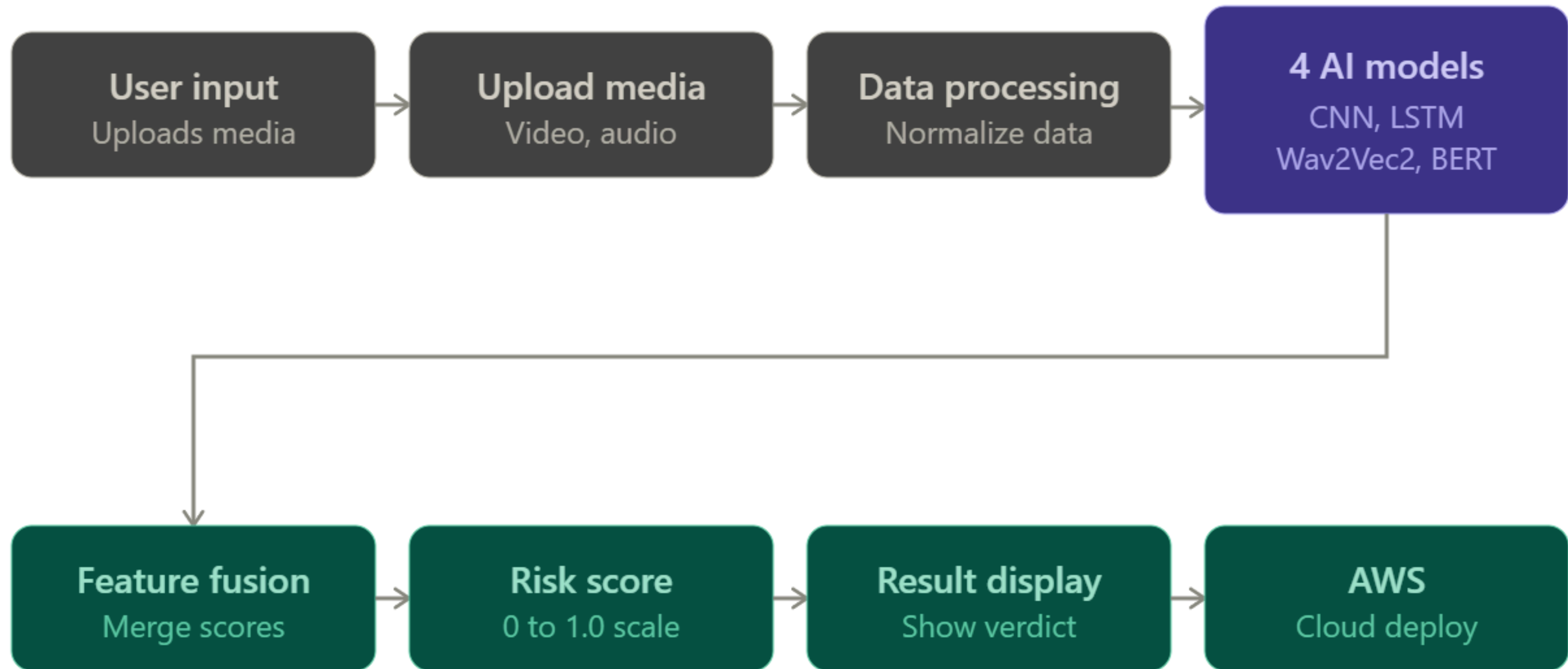
4

Risk Evaluation

A meta-classifier scores probability (0.0 to 1.0) and generates localized explainability highlights.



Flow Chart



Multi-Modal Fraud Intelligence

MULTI-MODAL ANALYSIS

Analyze Image, Video, Audio & Text for modality-wise fraud signals.

CROSS-MODAL CONSISTENCY

Compare visual and audio signals to detect lip-speech mismatches.

RISK SCORE

Fuse multiple model predictions into one overall fraud risk score.

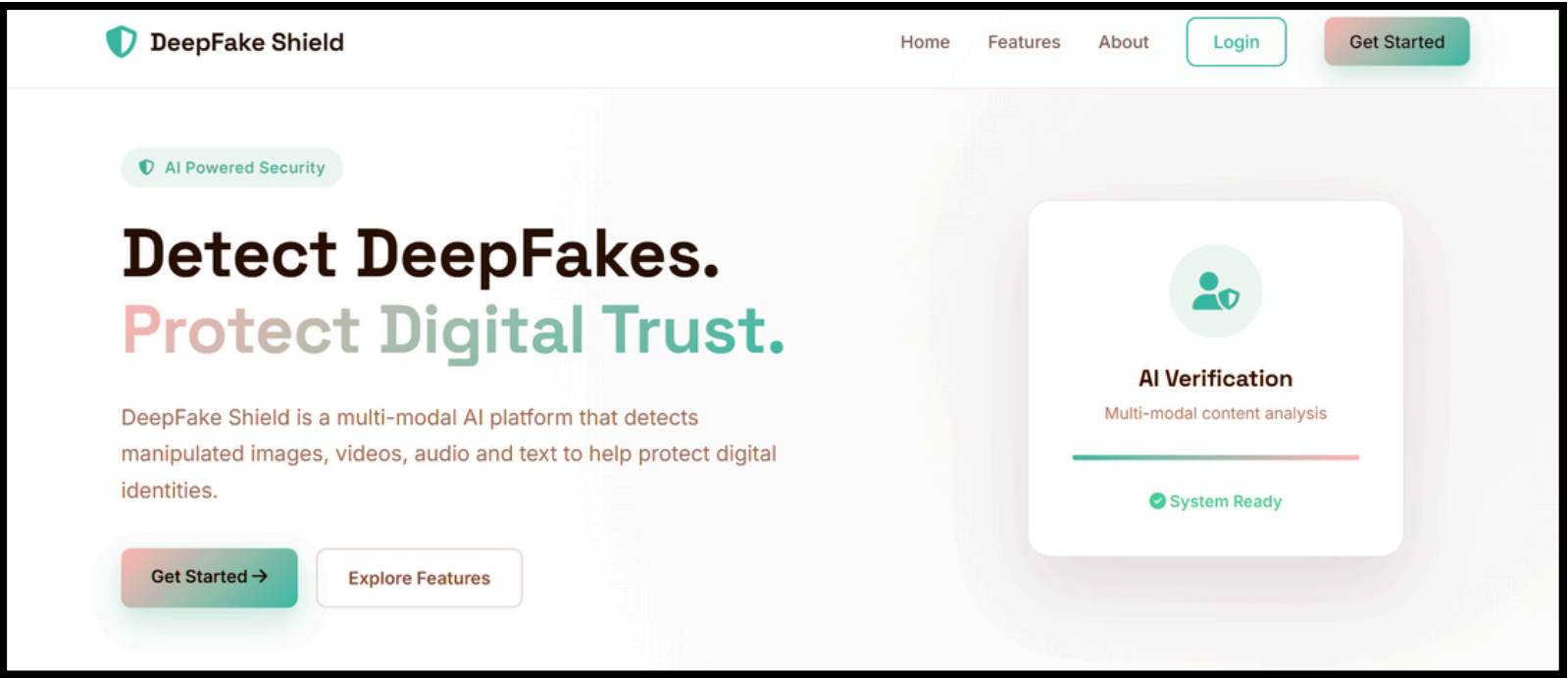
EXPLAINABLE DETECTION

Identify suspicious modalities and highlight key fraud indicators.

Work done

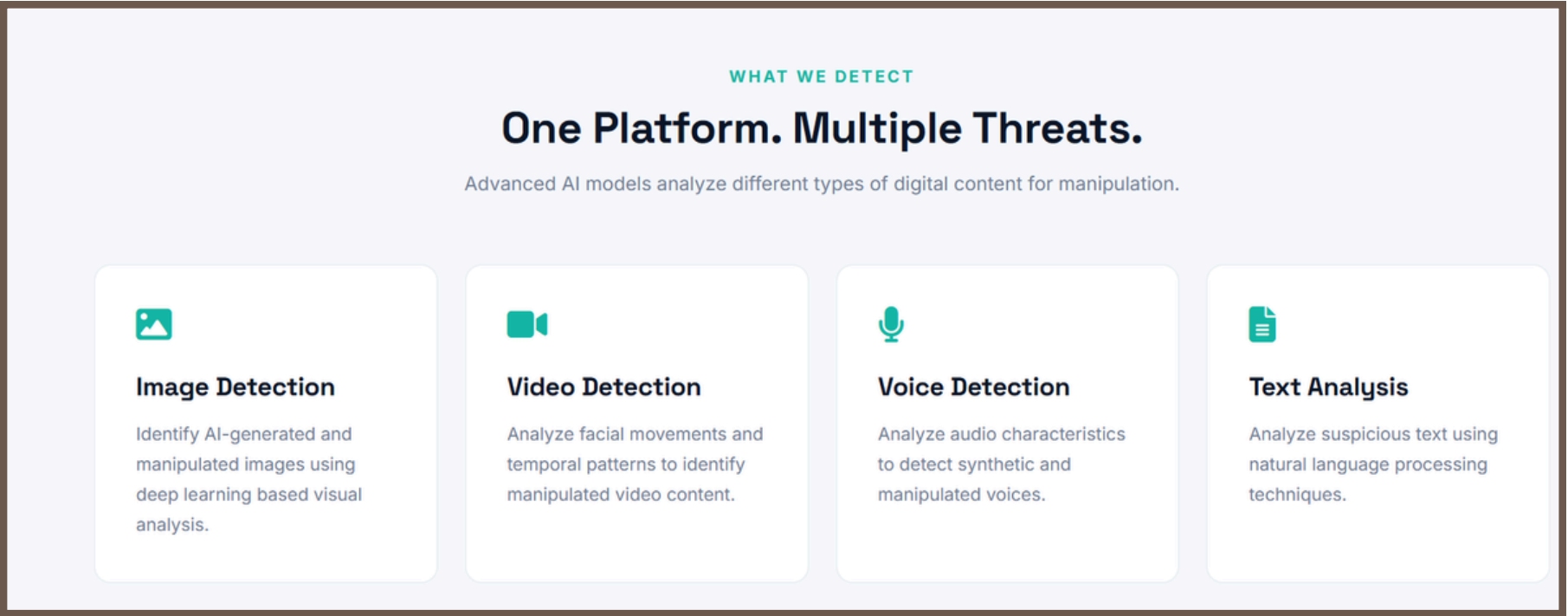
Initial Web Application Development :

Platform Overview



Home Page : Introduces the DeepFake-Shield platform and its purpose.

Core Capabilities

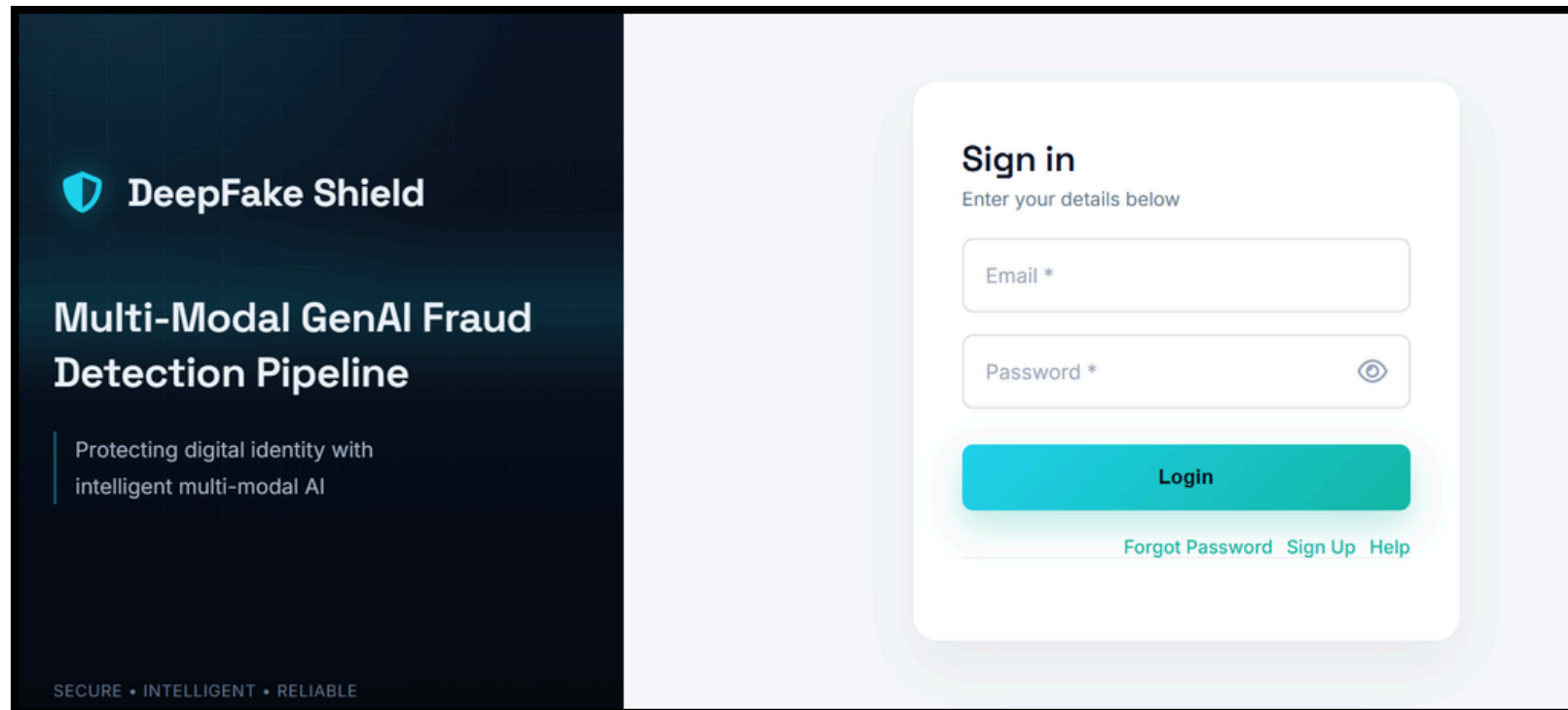


Features Page : Overview of the proposed multi-modal fraud detection capabilities

Work done

Initial Web Application Development :

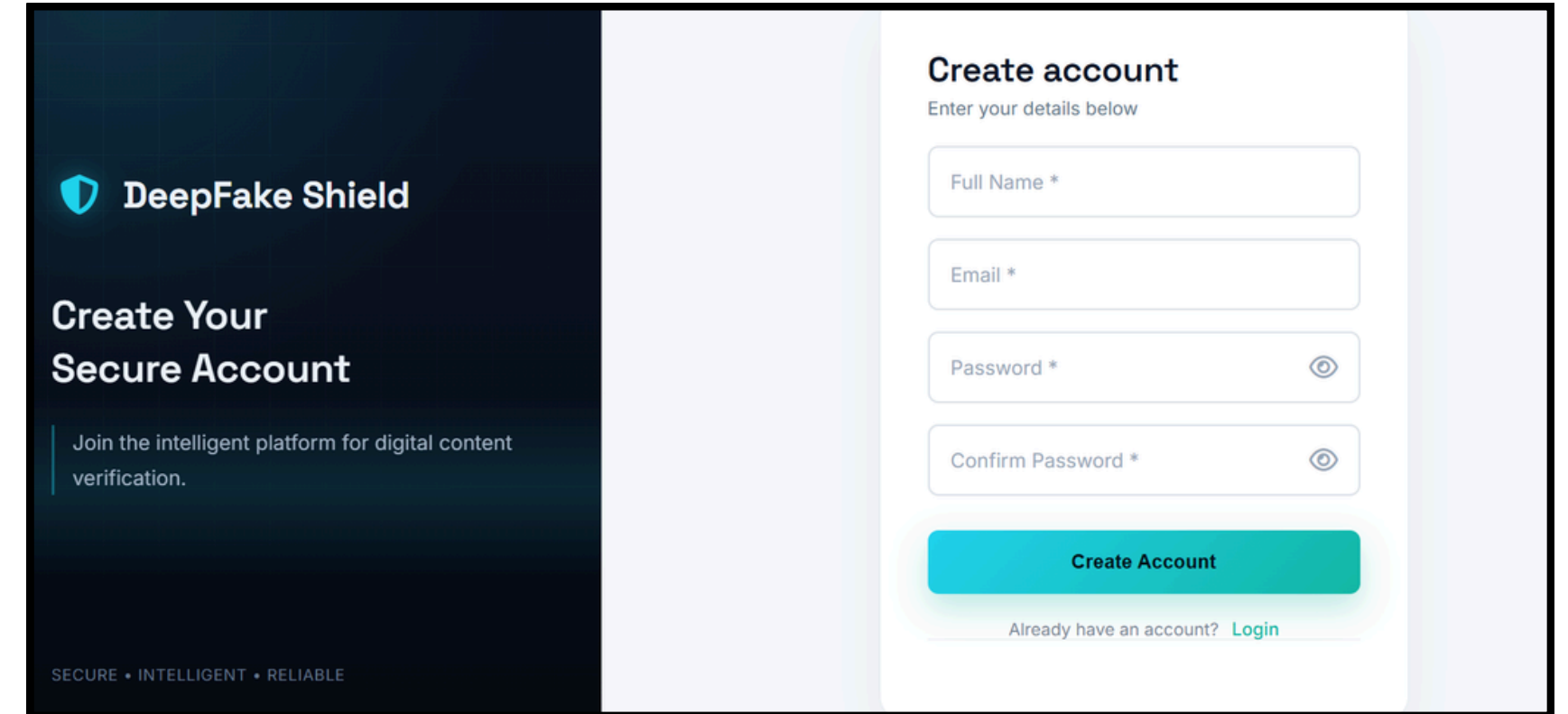
Secure User Access



The login page features a dark blue sidebar on the left with the 'DeepFake Shield' logo and the title 'Multi-Modal GenAI Fraud Detection Pipeline'. The main content area is light blue and contains a white 'Sign in' card. The card has fields for 'Email *' and 'Password *' with an eye icon for toggling visibility. A teal 'Login' button is at the bottom of the card, with links for 'Forgot Password', 'Sign Up', and 'Help' below it.

Login Page – Secure user authentication interface

New User Registration

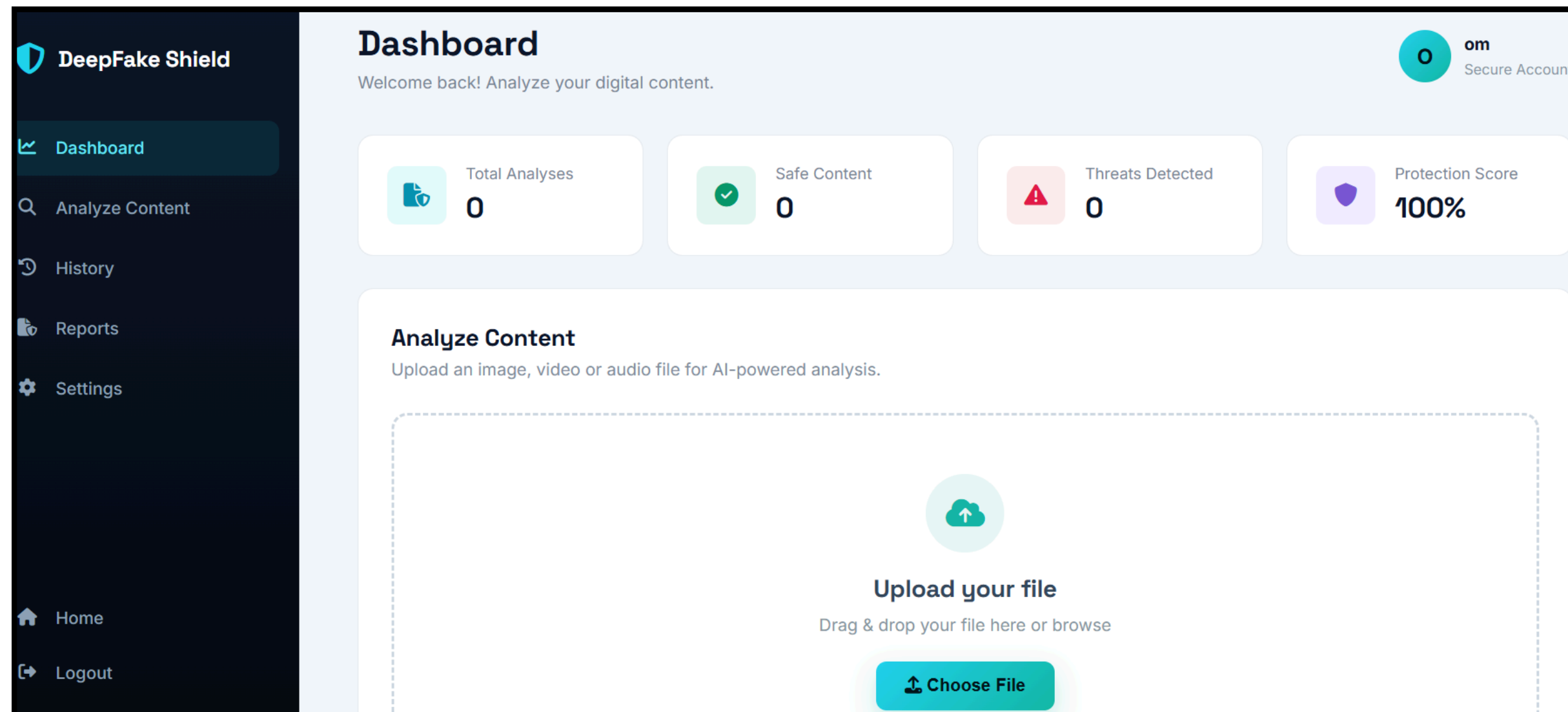


The registration page features a dark blue sidebar on the left with the 'DeepFake Shield' logo and the title 'Create Your Secure Account'. The main content area is light blue and contains a white 'Create account' card. The card has fields for 'Full Name *', 'Email *', 'Password *' (with an eye icon), and 'Confirm Password *' (with an eye icon). A teal 'Create Account' button is at the bottom of the card, with a link for 'Already have an account? Login' below it.

Registration Page – New user account creation interface

Work done

Initial Web Application Development :



DASHBOARD : Initial interface for accessing system functionality

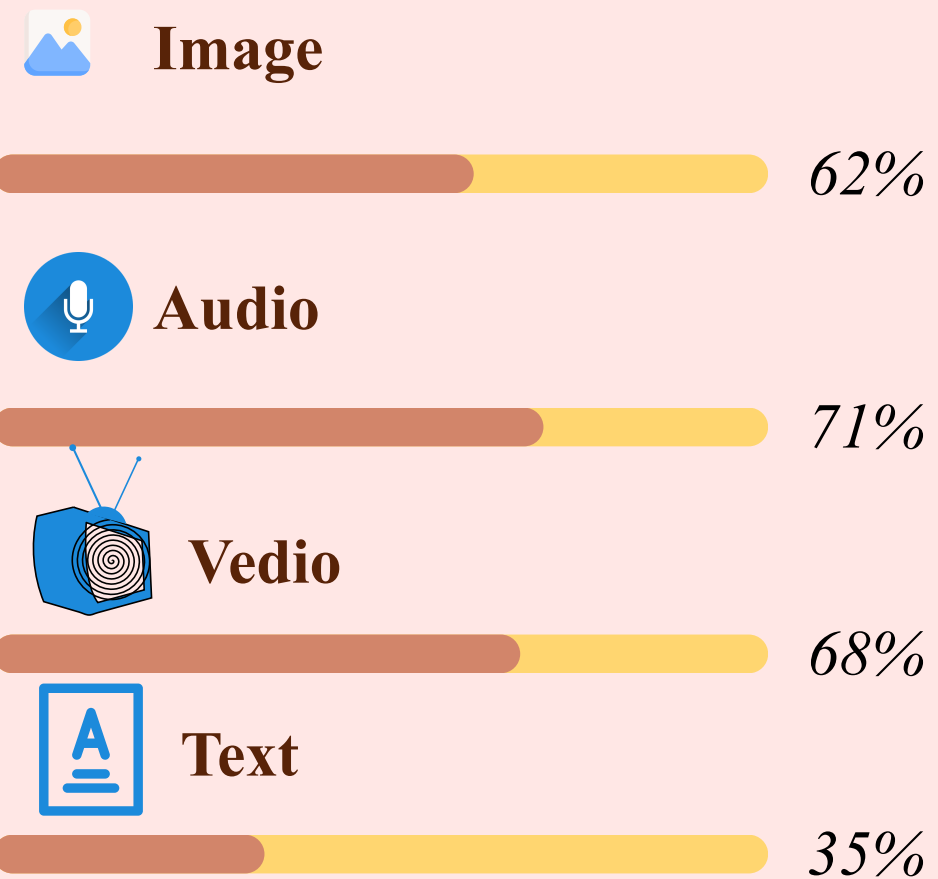
Work done (till date)

Completed the initial phase of the DeepFake-Shield project by conducting research, designing the system architecture, selecting technologies and developed the initial web application development.

- Conducted a literature survey of recent DeepFake Detection and Generative AI Fraud Detection research (2020–2025).
- Finalized the project technology stack: Python, Flask, HTML, CSS, JavaScript, TensorFlow, OpenCV, and AWS Cloud.
- Designed the complete software architecture and project workflow.
- Identified suitable datasets and selected AI models for image, video, audio, and text analysis.
- Designed the Multi-Modal Fraud Detection Pipeline.
- Prepared the system block diagram, project documentation, and presentation for the initial phase review.
- Started implementation of the DeepFake-Shield web application.
- Developed the Home Page and Features Page.
- Developed the Login and Registration interfaces.
- Developed the initial Dashboard interface.
- Implemented the basic Flask backend structure for the web application.

Result & Discussion)

MODALITY ANALYSIS



CROSS-MODAL CHECK



MISMATCH
Lip ↔ Speech



86%
FRAUD RISK

Illustrative output — actual scores will be generated after model integration and experimental testing.

Work – Plan

Months Activities	JUL'26	AUG'26	SEPT'26	OCT'26	NOV'26
Literature Reviews	√	√			
Component Identification & Selection		√			
Designing		√	√		
Fabrication		√	√	√	
Experimental Analysis		√	√	√	
Testing and Debugging		√	√	√	√
Preparation of Project Report			√	√	√
Thesis and Poster Submission				√	√

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- React Documentation – <https://react.dev>
- AWS Documentation – <https://aws.amazon.com/documentation/>

Thank you !